1 Introduction

Consider an environment where an agent must select an action from a set of available actions. The utility of each action for the agent can be expressed as a real-valued function, which depends on the unknown state of the environment. Specifically, let a denote an action, and x represent the unknown environment state. The utility function $U(a, x)$ quantifies the desirability of each action. Additionally, we assume that an evidence e of the environment state exists, and our agent can leverage this evidence to determine the optimal action.

According to Bayesian decision theory, the optimal action, when evidence e is observed, is the one that maximizes the expected utility conditioned on e:

$$a_{\text{optimal}} = \arg\max_{a} \mathbb{E}_{P^*}\left[U(a, x) \mid e\right] = \arg\max_{a} \sum_{x} U(a, x) P^*(x \mid e).$$

Here $P^*(X \mid e)$ is the probability distribution of the environment state given the evidence. However, how can we obtain this distribution? Is there an objective distribution inherent in the environment that we need to somehow discover?

From the subjective perspective on probability, the answer is negative. There is no such objective distribution, and P^* exists only in our minds. It serves as a tool that helps us regulate our reasoning—an intermediate measure created within the Bayesian decision-making framework to ensure we stay consistent with our assumptions and common sense principles.

Considering the subjective viewpoint, the determination of P^* relies on one's prior knowledge and assumptions. When we have no assumptions and no prior knowledge, the set of candidate distributions for P^* simply consists of all possible probability distributions. As we make assumptions or obtain some prior knowledge, the probability distributions that do not align with our assumptions or knowledge are eliminated from this set. Consequently, the set of acceptable probability distributions becomes more refined. We refer to this refined set as a probabilistic model:

Definition 1 (Probabilistic Model)**.** We define a *probabilistic model* $\mathcal{M}$ as a pair $(\mathcal{S}, \mathcal{P})$, where $\mathcal{S}$ is a sample space, and $\mathcal{P}$ is a set of acceptable probability measures on $\mathcal{S}$.

One of the most commonly used assumptions in the statistical modeling is the exchangeability assumption. We say a sequence of random variables $X_1, X_2, \ldots$ is exchangeable if the ordering of variables does not provide any useful information. In other words, for an exchangeable sequence of random variables, the joint probability is the same for any permutation of variables. This property holds true for every subset of the sequence.

For instance, if each X_i represents a binary variable, the joint probability depends solely on the total count of 1s and 0s, and the specific order in which these variables appear becomes irrelevant.

When a sequence of random variables is exchangeable, de Finetti's theorem states that their joint probability has a special representation:

$$P(x_1, x_2, \dots) = \int \prod_i \pi(x_i \mid \theta) p(\theta) d\theta.$$

This means, exchangeable observations are conditionally independent relative to some latent parameter θ.

Moreover, the theorem states that if each X_i is discrete, the distribution $\pi(x_i \mid \theta)$ corresponds to the multinomial distribution.

Now, consider our decision-making problem: Suppose the evidence e consists of a sequence of previous experiences of the environment. If we assume that our environment remains unchanged over time (making this sequence exchangeable), then to fully determine P^*, we only need to establish the prior probability of our latent parameter $p(\theta)$.

Definition 2 (Parametric Model)**.** Let Θ be a random vector. A *parametric model* is a probabilistic model $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$, where for every $P \in \mathcal{P}$ there is a prior probability distribution p, such that for every $s \in \mathcal{S}$

$$P(s) = \int \pi(s \mid \theta) p(\theta) d\theta, \tag{1}$$

while π is a known probability distribution.

Clearly, one approach to selecting a suitable prior distribution $p(\theta)$ is to rely on prior knowledge about the environment. However, what if such knowledge is unavailable?

It is not difficult to identify some distributions that are not good choices for $p(\theta)$. For instance, any distribution that assigns zero probabilities to certain values of θ is not a good choice. If we use such a prior, no evidence can convince us that the value of θ lies within those excluded values. Unfortunately, arguments like this one can only rule out a small portion of the potential distributions for $p(\theta)$, leaving us with numerous options and an unclear path for selection.

In this discussion, we will explore a new approach for choosing an appropriate prior distribution when prior knowledge about the environment is lacking.

2 Central Distribution

Assume that in an environment we are interested in the probability of a random variable X conditioned on an observed evidence e. We have decided on a probabilistic model $\mathcal{M}$ for the environment, and without any particular prior knowledge, we have chosen a distribution P from our model.

We ask an expert—a person familiar with the environment—to evaluate our distribution P. Presumably, that expert has his own beliefs about the environment and thinks we should use a different distribution Q, instead of P. Consequently, when evaluating P, he would try to measure the loss of using the incorrect distribution $P(X \mid e)$, while believing that the correct distribution is Q.

Let's assume that a real number can be used to measure this loss, and denote it as $\mathbb{D}\left(Q_{X|e} \parallel P_{X|e}\right)$. It seems reasonable as P gets closer to Q this measure decreases, and increases when P diverges from Q. Therefore, $\mathbb{D}\left(Q \parallel P\right)$ effectively is a measure of the divergence of P from Q, when Q is the true distribution.

Interestingly, if we make certain assumptions about the properties of this divergence measure, it turns out that—up to a constant factor—it must be equal to the Kullback–Leibler (KL) divergence. Alternatively, by adopting a more minimal set of properties, we can prove that if we accept Shannon's entropy as a measure of the uncertainty of a distribution, we should also consider KL divergence as the appropriate method for measuring divergence between distributions.

Theorem 1. *Suppose that $\mathbb{D}\left(Q \parallel P\right)$ is a divergence measure, defined for probability distributions Q and P, which satisfies the following properties:*

Consistency with entropy *If U is the uniform distribution with the same support as Q, then $\mathbb{D}\left(Q \parallel U\right) = \mathbb{H}\left[U\right] - \mathbb{H}\left[Q\right]$, where $\mathbb{H}$ denotes Shannon's entropy.*

Additivity *For any two random variables X, Y*

$$\mathbb{D}\left(Q_{X,Y} \parallel P_{X,Y}\right) = \mathbb{D}\left(Q_X \parallel P_X\right) + \sum_x \mathbb{D}\left(Q_{Y|x} \parallel P_{Y|x}\right) Q(x)\cdot$$

Then, $\mathbb{D}\left(Q \parallel P\right)$ is the Kullback-Leibler divergence.

Proof. Let X be a discrete random variable, and $P(X)$ and $Q(X)$ be two arbitrary distributions for X. We define a new continuous random variable Ω and define

$$p(\Omega = \omega \mid X = x) = \begin{cases} \frac{1}{P(x)} & 0 \leq \omega \leq P(X = x), \\ 0 & \text{otherwise}\cdot \end{cases}$$

Notice that both $P(\Omega \mid X)$ and $P(\Omega, X)$ are uniform distributions, and $\mathbb{H}\left[P_{\Omega|x}\right] = \ln P(x)$, $\mathbb{H}\left[P_{\Omega,X}\right] = 0$.

We define $Q(\Omega \mid x)$ to be an arbitrary distribution that has the same support as $P(\Omega \mid x)$. This ensures that $Q(\Omega, X)$ and $P(\Omega, X)$ have the same support as well. Thus, we have

$$\begin{aligned}\mathbb{D}\left(Q_{\Omega,X} \parallel P_{\Omega,X}\right) &= \mathbb{H}\left[P_{\Omega,X}\right] - \mathbb{H}\left[Q_{\Omega,X}\right] \\ &= \sum_x \int q(\omega, x) \ln q(\omega, x) d\omega \\ &= \sum_x Q(x) \int q(\omega \mid x) \ln q(\omega, x) d\omega,\end{aligned}$$

and

$$\begin{aligned}\mathbb{D}\left(Q_{\Omega|x} \parallel P_{\Omega|x}\right) &= \mathbb{H}\left[P_{\Omega|x}\right] - \mathbb{H}\left[Q_{\Omega|x}\right] \\ &= \ln P(x) + \int q(\omega \mid x) \ln q(\omega \mid x) d\omega \cdot\end{aligned}$$

Now, we use the additivity property to obtain a formula for our divergence measure:

$$\begin{aligned}\mathbb{D}\left(Q_X \parallel P_X\right) &= \mathbb{D}\left(Q_{\Omega,X} \parallel P_{\Omega,X}\right) - \sum_x \mathbb{D}\left(Q_{\Omega|x} \parallel P_{\Omega|x}\right) Q(x) \\ &= \sum_x Q(x) \left(\int q(\omega \mid x) \ln \frac{q(\omega, x)}{q(\omega \mid x)} d\omega - \ln P(x)\right) \\ &= \sum_x Q(x) \left(\ln Q(x) \int q(\omega \mid x) d\omega - \ln P(x)\right) \\ &= \sum_x Q(x) \ln \frac{Q(x)}{P(x)} \cdot\end{aligned}$$

□

In this theorem, the first property simply states that the entropy of a distribution is related to the divergence of the uniform distribution. The more the divergence is, the less the entropy should be.

The second property tells us that the divergence is an additive quantity. We know that the divergence of $P(X, Y)$ from $Q(X, Y)$ is actually a measure of the generic loss of using $P(X, Y)$ when we believe $Q(X, Y)$ is the true distribution. To calculate this loss, we can assume that X happens before Y. In that way, this loss would include the loss of using $P(X)$, and then, if we observe $X = x$, which can happen with probability $Q(x)$, it would also include the loss of using $P(Y \mid x)$. This property indicates that for combining these individual terms we should use addition.

Returning to our earlier discussion, this implies that to evaluate P our expert should use the KL divergence between P and Q. What if we find a distribution C that has a small KL divergence from any other distribution in our model? In that way, every expert would agree that our distribution is a good choice. Interestingly, this relates to the concept of Bayesian convergence, where the posterior probability converges to the same distribution regardless of the prior distribution.

Definition 3 (Query Set)**.** A *query set* is a set of pairs of discrete random variables, all defined on the same sample space.

Definition 4. Let $\mathcal{X} = \{(X_1, E_1), (X_2, E_2), \ldots\}$, be a query set in a probabilistic model $\mathcal{M} \equiv (\mathcal{S}, \mathcal{P})$. We define the *divergence measure of $P \in \mathcal{P}$ for $\mathcal{X}$* as follows:

$$\mathbb{D}_{\mathcal{X}}[P] = \max_{Q \in \mathcal{P}} \mathbb{E}_Q \left[\sum_i \mathbb{D}\left(Q_{X_i|e_i} \parallel P_{X_i|e_i}\right) \right], \tag{2}$$

where $\mathbb{E}_Q[\cdot]$ denotes expectation relative to Q and each $\mathbb{D}\left(Q_{X_i|e_i} \parallel P_{X_i|e_i}\right)$ is the Kullback-Leibler divergence between $P(X_i \mid e_i)$ and $Q(X_i \mid e_i)$:

$$\mathbb{D}\left(Q_{X_i|e_i} \parallel P_{X_i|e_i}\right) = \sum_{x_i} Q(x_i \mid e_i) \ln \frac{Q(x_i \mid e_i)}{P(x_i \mid e_i)}.$$

Proposition. Using the linearity of expectation, we have

$$\mathbb{D}_{\mathcal{X}}[P] = \max_{Q \in \mathcal{P}} \sum_i \sum_{x_i, e_i} Q(x_i, e_i) \ln \frac{Q(x_i \mid e_i)}{P(x_i \mid e_i)}.$$

In this definition, the query set $\mathcal{X} = \{(X_1, E_1), (X_2, E_2), \ldots\}$ plays a crucial role. Specifically, $\mathcal{X}$ defines the set of random variables for which we are interested in their distributions. Each X_i represents a random variable whose distribution is important to us, and each E_i corresponds to the evidence that is available for that variable.

For a parametric model (as defined in Definition 2), it seems reasonable to define a conditional form of our divergence measure, conditioned on the value of the parameter.

To do so, in Equation 2, we replace normal expectation with conditional expectation. There will be no longer a need to maximize over different distributions, since in a parametric model the parameter fully determines the distribution of the sample space.

Definition 5 (Conditional Divergence Measure)**.** Assume that $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$ is a parametric model, and $\mathcal{X} = \{(X_1, E_1), (X_2, E_2), \ldots\}$ is a query set. We define the *divergence measure of* $P \in \mathcal{P}$ *given* θ *for* $\mathcal{X}$ as follows:

$$\mathbb{D}_\mathcal{X}[P \mid \theta] = \mathbb{E}_\pi \left[\sum_i \mathbb{D}\left(\pi_{X_i|e_i,\theta} \parallel P_{X_i|e_i}\right) \middle| \theta \right] \tag{3}$$

$$= \sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{P(x_i \mid e_i)} \cdot \tag{4}$$

Even if we cannot find a distribution that has a small divergence from every distribution, still using the distribution that minimizes the maximum divergence seems a reasonable choice.

Definition 6 (Central Distribution)**.** In a model $\mathcal{M} \equiv (\mathcal{S}, \mathcal{P})$, we define the *central distribution* for a query set $\mathcal{X}$ to be a distribution $C_\mathcal{X} \in \mathcal{P}$ which minimizes the divergence measure for $\mathcal{X}$:

$$C_\mathcal{X} = \arg\min_{P \in \mathcal{P}} \mathbb{D}_\mathcal{X}[P] \cdot \tag{5}$$

In a parametric model, the maximum of divergence happens for a prior distribution that is concentrated at a single value of the parameter θ. That means, for a parametric model which allows any high peak prior, instead of maximizing over the set of all acceptable distributions, we can simply maximize over the parameter space, using the conditional version of the divergence measure.

Lemma 1. *Assume that* $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$ *is a parametric model. Let* $\mathcal{X} = \{(X_1, E_1), (X_2, E_2), \ldots\}$ *be a query set in* $\mathcal{M}_\Theta$*. For every* $P \in \mathcal{P}$ *we have*

$$\mathbb{D}_\mathcal{X}[P] \leq \max_{\theta \in \Theta} \mathbb{D}_\mathcal{X}[P \mid \theta]$$

Proof. Let R be an arbitrary distribution in $\mathcal{M}_\Theta$, since the Kullback-Leibler divergence is positive, its expected value is also positive, and for every parameter value θ we have

$$\sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{R(x_i \mid e_i)} \geq 0 \cdot$$

Using this, for every distribution P and every θ, we obtain

$$\sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{R(x_i \mid e_i)}{P(x_i \mid e_i)} \leq \mathbb{D}_\mathcal{X}[P \mid \theta] \cdot$$

Since this inequality holds for any θ, the maximum of the right-hand side w.r.t. θ is also greater than or equal to the maximum of the left-hand side:

$$\max_{\theta\in\Theta}\sum_i\sum_{x_i,e_i}\pi(x_i,e_i\mid\theta)\ln\frac{R(x_i\mid e_i)}{P(x_i\mid e_i)}\leq\max_{\theta\in\Theta}\mathbb{D}_{\mathcal{X}}\left[P\mid\theta\right]\cdot \tag{6}$$

$\mathcal{M}_\Theta$ is a parametric model and $R(x_i,e_i)=\int\pi(x_i,e_i\mid\theta)r(\theta)d\theta$, where $r(\theta)$ is a prior probability distribution. Trivially, for every θ we have

$$\sum_i\sum_{x_i,e_i}\pi(x_i,e_i\mid\theta)\ln\frac{R(x_i\mid e_i)}{P(x_i\mid e_i)}\leq\max_{\theta\in\Theta}\sum_i\sum_{x_i,e_i}\pi(x_i,e_i\mid\theta)\ln\frac{R(x_i\mid e_i)}{P(x_i\mid e_i)}\cdot$$

Since r is a distribution and is non-negative, we can multiply this inequality by $r(\theta)$ and integrate. Notice that the right hand side is not a function of θ and it remains unchanged because $\int r(\theta)d\theta=1$:

$$\mathbb{E}_R\left[\sum_i\mathbb{D}\left(R_{X_i|e_i}\parallel P_{X_i|e_i}\right)\right]\leq\max_{\theta\in\Theta}\sum_i\sum_{x_i,e_i}\pi(x_i,e_i\mid\theta)\ln\frac{R(x_i\mid e_i)}{P(x_i\mid e_i)}\cdot$$

Now from Inequality 6, for *any* distribution P and R in $\mathcal{M}_\Theta$, it follows that

$$\mathbb{E}_R\left[\sum_i\mathbb{D}\left(R_{X_i|e_i}\parallel P_{X_i|e_i}\right)\right]\leq\max_{\theta\in\Theta}\mathbb{D}_{\mathcal{X}}\left[P\mid\theta\right],$$

which proves the lemma. □

Theorem 2. *Let $\mathcal{M}_\Theta\equiv(\mathcal{S},\mathcal{P})$ be a parametric model, and $\mathcal{X}=\{(X_1,E_1),(X_2,E_2),\dots\}$ be a query set in $\mathcal{M}_\Theta$. If for every value of θ there is a distribution $R\in\mathcal{P}$ such that $R(s)=\pi(s\mid\theta)$ for every $s\in\mathcal{S}$, then for any $P\in\mathcal{P}$ we have:*

$$\mathbb{D}_{\mathcal{X}}\left[P\right]=\max_{\theta\in\Theta}\mathbb{D}_{\mathcal{X}}\left[P\mid\theta\right]\cdot \tag{7}$$

Proof. For an arbitrary distribution $P\in\mathcal{P}$, we let

$$\theta^*=\arg\max_{\theta\in\Theta}\mathbb{D}_{\mathcal{X}}\left[P\mid\theta\right]\cdot$$

In $\mathcal{M}_\Theta$ there should be a distribution $R\in\mathcal{P}$ that equals $\pi(s\mid\theta^*)$ for every $s\in\mathcal{S}$, and therefore it defines the same distribution for any random variable:

$$\begin{aligned}\mathbb{E}_R\left[\sum_i\mathbb{D}\left(R_{X_i|e_i}\parallel P_{X_i|e_i}\right)\right]&=\mathbb{E}_\pi\left[\sum_i\mathbb{D}\left(\pi_{X_i|e_i,\theta^*}\parallel P_{X_i|e_i}\right)\middle|\,\theta^*\right]\\&=\max_{\theta\in\Theta}\mathbb{D}_{\mathcal{X}}\left[P\mid\theta\right]\cdot\end{aligned}$$

On the other hand, clearly

$$\max_{Q\in\mathcal{P}} \mathbb{E}_Q\left[\sum_i \mathbb{D}\left(Q_{X_i|e_i} \parallel P_{X_i|e_i}\right)\right] \geq \mathbb{E}_R\left[\sum_i \mathbb{D}\left(R_{X_i|e_i} \parallel P_{X_i|e_i}\right)\right]\cdot$$

Notice that the left hand side is $\mathbb{D}_{\mathcal{X}}[P]$, and

$$\mathbb{D}_{\mathcal{X}}[P] \geq \max_{\theta\in\Theta} \mathbb{D}_{\mathcal{X}}[P \mid \theta]\cdot$$

This inequality and Lemma 1 prove the theorem. □

Definition 7 (Generic Parametric Model)**.** We say a parametric model $\mathcal{M}_\Theta \equiv (\mathcal{S},\mathcal{P})$ is *generic*, if for any arbitrary probability distribution r, there is a probability measure $R \in \mathcal{P}$, such that for every $s \in \mathcal{S}$

$$R(s) = \int \pi(s \mid \theta) r(\theta) d\theta\cdot$$

Proposition. A generic parametric model satisfies the conditions of Theorem 2.

Theorem 3 (Minimax theorem [2])**.** *Let $\mathcal{X} \subset \mathbb{R}^n$ and $\mathcal{Y} \subset \mathbb{R}^m$ be compact convex sets. If $f : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$ is a continuous function that is concave-convex, i.e.*

- *$f(\cdot, y) : \mathcal{X} \to \mathbb{R}$ is concave for fixed y, and*
- *$f(x, \cdot) : \mathcal{Y} \to \mathbb{R}$ is convex for fixed x.*

Then we have that

$$\max_{x\in\mathcal{X}} \min_{y\in\mathcal{Y}} f(x, y) = \min_{y\in\mathcal{Y}} \max_{x\in\mathcal{X}} f(x, y)\cdot \tag{8}$$

Theorem 4. *Consider a generic parametric model $\mathcal{M}_\Theta \equiv (\mathcal{S},\mathcal{P})$, and let $\mathcal{X} = \{(X_1, E_1), (X_2, E_2), \ldots\}$ be a set of pairs of discrete random variables in $\mathcal{M}_\Theta$. Assume that for every $Q_1, Q_2 \in \mathcal{P}$ and any real number $0 \leq \alpha \leq 1$, there is a distribution $Q^* \in \mathcal{P}$, such that for every i, x_i, e_i we have*

$$Q^*(x_i \mid e_i) = \alpha Q_1(x_i \mid e_i) + (1-\alpha) Q_2(x_i \mid e_i)\cdot \tag{9}$$

In this model, the central distribution for $\mathcal{X}$ can be obtained by:

$$C_{\mathcal{X}} = \arg\max_{P\in\mathcal{P}} \int \mathbb{D}_{\mathcal{X}}[P \mid \theta]\, p(\theta) d\theta \tag{10}$$

$$= \arg\max_{P\in\mathcal{P}} \int \sum_i \sum_{x_i, e_i} p(x_i, e_i, \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{P(x_i \mid e_i)} d\theta, \tag{11}$$

where $P(s) = \int \pi(s \mid \theta) p(\theta) d\theta$, for every $s \in \mathcal{S}$.

Proof. We prove this theorem using the Minimax theorem. For that, we define a functional as follows:

$$f(r, P_{X_i|E_i}) = \int \mathbb{E}_\pi \left[\sum_i \mathbb{D}\left(\pi_{X_i|e_i,\theta} \parallel P_{X_i|e_i}\right) \middle| \theta \right] r(\theta) d\theta,$$

where $r(\theta)$ is an arbitrary probability distribution, and $P \in \mathcal{P}$. Since $r(\theta)$ can be any distribution, maximizing with respect to r is like maximizing with respect to θ, and we have

$$C_{\mathcal{X}} = \arg \min_{P \in \mathcal{P}} \max_r f(r, P_{X_i|E_i})\cdot$$

The domain of r is the set of all probability distributions, which is a compact convex set. Moreover, from Equation 9, it follows that the set of probability distributions for $X_i|E_i$ in $\mathcal{M}_\Theta$ is also a compact convex set. Thus, the domain of $f(r, P_{X_i|E_i})$ is a compact convex set.

For a fixed $P_{X_i|E_i}$, $f(\cdot, P_{X_i|E_i})$ is a linear function, which is both concave and convex. On the other hand, if we let $R(x_i, e_i) = \int \pi(x_i, e_i \mid \theta) r(\theta) d\theta$ for $f(r, \cdot)$ we have

$$f(r, \cdot) = c - \sum_i \sum_{x_i, e_i} R(x_i, e_i) \ln P(x_i \mid e_i)\cdot \tag{12}$$

If we calculate the Hessian matrix of $f(r, \cdot)$, we observe that the matrix is diagonal, and each element of the main diagonal is $\frac{R(x_i|e_i)}{P(x_i|e_i)^2}$. Thus, the hessian matrix of $f(\cdot, P_{X_i|E_i})$ is positive semi-definite and $f(r, \cdot)$ is a convex function. Now, Theorem 3 implies

$$\min_{P \in \mathcal{P}} \max_r f(r, P_{X_i|E_i}) = \max_r \min_{P \in \mathcal{P}} f(r, P_{X_i|E_i})\cdot \tag{13}$$

From Equation 12 we can see that for an arbitrary fixed r we have

$$\min_{P \in \mathcal{P}} f(r, P_{X_i|E_i}) = \max_{P \in \mathcal{P}} \sum_i \sum_{x_i, e_i} R(x_i, e_i) \ln P(x_i \mid e_i)\cdot$$

We know that the Kullback–Leibler divergence, and therefore its expected value, is non-negative:

$$\mathbb{E}_R \left[\sum_i \mathbb{D}\left(R_{X_i|e_i} \parallel P_{X_i|e_i}\right) \right] \geq 0\cdot$$

By simple algebra we get

$$\sum_i \sum_{x_i, e_i} R(x_i, e_i) \ln P(x_i \mid e_i) \leq \sum_i \sum_{x_i, e_i} R(x_i, e_i) \ln R(x_i \mid e_i)\cdot$$

Recall that $\mathcal{M}_\Theta$ is a generic parametric model, so $R \in \mathcal{P}$, and we can reach this upper bound by letting $P = R$:

$$\min_{P\in\mathcal{P}} f(r, P_{X_i|E_i}) = \int \sum_i \sum_{x_i,e_i} r(x_i, e_i, \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{R(x_i \mid e_i)} d\theta\cdot$$

By Equation 13 we conclude

$$C_{\mathcal{X}} = \arg\max_{R\in\mathcal{R}} \int \mathbb{E}_\pi \left[\sum_i \mathbb{D}\left(\pi_{X_i|e_i,\theta} \parallel R_{X_i|e_i}\right) \middle| \theta \right] r(\theta) d\theta\cdot \qquad \square$$

Theorem 5. *Under the assumptions of Theorem 4, the central distribution for $\mathcal{X}$ satisfies:*

$$\mathbb{D}_{\mathcal{X}}\left[C_{\mathcal{X}} \mid \theta\right] = \lambda, \tag{14}$$

where λ is a constant.

Proof. By Theorem 4, the central distribution for $\mathcal{X}$ maximizes

$$\mathcal{F}\left[p(\theta)\right] = \int \sum_i \sum_{x_i,e_i} p(x_i, e_i, \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{P(x_i \mid e_i)} d\theta,$$

with respect to different choices of the prior distribution $p(\theta)$, subject to the constraint $\int p(\theta)d\theta = 1$.

Applying the Lagrange multiplier technique, the local maximum of $\mathcal{F}$ is a stationary point of the Lagrangian functional:

$$\mathcal{L}\left[p(\theta), \lambda\right] = \int \sum_i \sum_{x_i,e_i} p(x_i, e_i, \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{P(x_i \mid e_i)} d\theta - \lambda \left(\int p(\theta) d\theta - 1 \right)\cdot$$

Let us denote the prior distribution corresponding to the central distribution $C_{\mathcal{X}}$ by $c_{\mathcal{X}}(\theta)$. Let $p(\theta) = c_{\mathcal{X}}(\theta) + \epsilon\eta(\theta)$ be a variation of $c_{\mathcal{X}}(\theta)$, where $\eta(\theta)$ is a continuously differentiable function with $\eta(0) = \eta(1) = 0$, and ϵ is a small constant.

Since $c_{\mathcal{X}}(\theta)$ is an extremizing function for $\mathcal{F}\left[p(\theta)\right]$, the lagrangian $\mathcal{L}\left[\epsilon, \lambda\right]$ has a stationary point where $\epsilon = 0$, and we have

$$\left.\frac{\partial \mathcal{L}\left[\epsilon, \lambda\right]}{\partial \epsilon}\right|_{\epsilon=0} = 0\cdot$$

Note that

$$\begin{aligned} \frac{dp(\theta)}{d\epsilon} &= \eta(\theta), \\ \frac{dP(x_i, e_i)}{d\epsilon} &= \int \pi(x_i, e_i \mid \theta)\eta(\theta) d\theta, \\ \frac{dP(e_i)}{d\epsilon} &= \int \pi(e_i \mid \theta)\eta(\theta) d\theta\cdot \end{aligned}$$

Using the chain rule, we calculate the partial derivative of $\mathcal{L}[\epsilon, \lambda]$ relative to ϵ:

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial \epsilon} &= \int \left(\sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{P(x_i \mid e_i)} \right) \eta(\theta) d\theta \\ &\quad - \int \left(\sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \frac{\int \pi(x_i, e_i \mid \theta)\eta(\theta) d\theta}{P(x_i, e_i)} \right) p(\theta) d\theta \\ &\quad + \int \left(\sum_i \sum_{e_i} \pi(e_i \mid \theta) \frac{\int \pi(e_i \mid \theta)\eta(\theta) d\theta}{P(e_i)} \right) p(\theta) d\theta \\ &\quad - \int \lambda \eta(\theta) d\theta\end{aligned}$$

Since $P(x_i, e_i) = \int \pi(x_i, e_i \mid \theta) p(\theta) d\theta$ and $P(e_i) = \int \pi(e_i \mid \theta) p(\theta) d\theta$ we get

$$\frac{\partial \mathcal{L}}{\partial \epsilon} = \int \left(-\lambda + \sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{P(x_i \mid e_i)} \right) \eta(\theta) d\theta \cdot$$

We let $\frac{\partial \mathcal{L}}{\partial \epsilon}\Big|_{\epsilon=0} = 0$. Since $\eta(\theta)$ is an arbitrary function, by applying the fundamental lemma of the calculus of variations and letting $\epsilon = 0$, for every value of θ we have

$$\sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{C_{\mathcal{X}}(x_i \mid e_i)} = \lambda \cdot \qquad \square$$

Consider a scenario where we are trying to predict the outcome of a four-sided dice. This dice is unfair, with each face marked by a binary number: 00, 01, 10, and 11. Before making our prediction, we can roll the dice for several times and use the results to learn the associated probabilities. We assume that the dice rolls are independent.

Let's focus on determining the central distribution for this problem. The first step in this process is to establish our query set $\mathcal{X} = \{(X_1, E_1), (X_2, E_2), \ldots\}$.

In a prediction problem, we are interested in the distribution of the random variable we want to predict, conditioned on the observed data. For example, if we roll the dice for three times before making our prediction, we will need $P(X_4 \mid X_1, X_2, X_3)$, where X_i denotes the outcome of the i-th dice roll.

Thus, one natural way for defining a query set $\mathcal{X} = \{(X_1, E_1), (X_2, E_2), \ldots\}$ is to let each X_i be the outcome of the i-th dice roll, and define each E_i as a vector of the previous dice outcomes $(X_1, X_2, \ldots, X_{i-1})$.

Now let us calculate the conditional divergence measure for $\mathcal{X}$ as defined in Definition 5:

$$
\begin{aligned}
\mathbb{D}_{\mathcal{X}}\left[P \mid \theta\right] = & \sum_{x_1} \pi(x_1 \mid \theta) \ln \frac{\pi(x_1 \mid \theta)}{P(x_1)} \\
& + \sum_{x_2, x_1} \pi(x_2, x_1 \mid \theta) \ln \frac{\pi\left(x_2 \mid x_1, \theta\right)}{P\left(x_2 \mid x_1\right)} \\
& + \sum_{x_3, x_1, x_2} \pi(x_3, x_1, x_2 \mid \theta) \ln \frac{\pi\left(x_3 \mid x_1, x_2, \theta\right)}{P\left(x_3 \mid x_1, x_2\right)} \\
& + \cdots \\
& + \sum_{x_n, x_1, \ldots, x_{n-1}} \pi(x_n, x_1, \ldots, x_{n-1} \mid \theta) \ln \frac{\pi\left(x_n \mid x_1, \ldots, x_{n-1}, \theta\right)}{P\left(x_n \mid x_1, \ldots, x_{n-1}\right)} \cdot
\end{aligned}
$$

After factoring out $\pi(x_1, \ldots, x_n \mid \theta)$ from every term, we use the chain rule for conditional probability:

$$
\begin{aligned}
\mathbb{D}_{\mathcal{X}}\left[P \mid \theta\right] &= \sum_{x_1, \ldots, x_n} \pi(x_1, \ldots, x_n \mid \theta) \ln \frac{\pi(x_1 \mid \theta)\pi(x_2 \mid x_1, \theta) \cdots \pi(x_n \mid x_1, \ldots, x_{n-1}, \theta)}{P(x_1)P(x_2 \mid x_1) \cdots P(x_n \mid x_1, \ldots, x_{n-1})} \\
&= \sum_{x_1, \ldots, x_n} \pi(x_1, \ldots, x_n \mid \theta) \ln \frac{\pi(x_1, \ldots, x_n \mid \theta)}{P(x_1, \ldots, x_n)} \cdot
\end{aligned}
$$

By Equation 11, for the central distribution we have

$$
\begin{aligned}
C_{\mathcal{X}} &= \arg\max_P \int \left(\sum_{x_1, \ldots, x_n} \pi(x_1, \ldots, x_n \mid \theta) \ln \frac{\pi(x_1, \ldots, x_n \mid \theta)}{P(x_1, \ldots, x_n)} \right) p(\theta) d\theta \\
&= \arg\max_P \int \sum_{x_1, \ldots, x_n} p(x_1, \ldots, x_n, \theta) \ln \frac{p(\theta \mid x_1, \ldots, x_n)}{p(\theta)} d\theta \cdot
\end{aligned}
$$

That implies, the central distribution for our problem is the same as the distribution we obtain when using the reference prior [1] for our main parameter.

Imagine playing a game of chance involving our four-sided dice. After rolling the dice, we input the outcome into a mysterious machine. The machine then announces whether we have won or lost the game. We have no clue how the machine works. We are not even sure if the dice outcome truly affects the machine's output. All we know is that the machine is memory-less, and the result of each game is independent.

Our goal is to learn how the machine operates through experimentation. We aim to be able to predict whether we will win the game after observing the dice outcome.

We assume that, first, we roll the dice n times, and record each outcome. Next, we input some of these dice outcomes into the machine and observe the result of each game. As we will see, this assumption is important for deriving the central distribution more easily.

Let us define a query set for this problem. We denote this set as $\mathcal{X} = \{(X_1, E_1), (X_2, E_2), \ldots\}$, where each X_i is the result of the i-th game, and the evidence E_i is a vector

$$(X_1, X_2, \ldots, X_{i-1}, Y_1, Y_2, \ldots, Y_n),$$

where each Y_i denotes the i-th dice outcome.

The conditional divergence for $\mathcal{X}$ can be obtained by

$$\begin{aligned}\mathbb{D}_{\mathcal{X}}\left[P \mid \theta\right] = & \sum_{x_1, y_1, \ldots, y_n} \pi(x_1, y_1, \ldots, y_n \mid \theta) \ln \frac{\pi\left(x_1 \mid y_1, \ldots, y_n, \theta\right)}{P\left(x_1 \mid y_1, \ldots, y_n\right)} \\ & + \sum_{x_2, x_1, y_1, \ldots, y_n} \pi(x_2, x_1, y_1, \ldots, y_n \mid \theta) \ln \frac{\pi\left(x_2 \mid x_1, y_1, \ldots, y_n, \theta\right)}{P\left(x_2 \mid x_1, y_1, \ldots, y_n\right)} \\ & + \sum_{x_3, x_1, x_2, y_1, \ldots, y_n} \pi(x_3, x_1, x_2, y_1, \ldots, y_n \mid \theta) \ln \frac{\pi\left(x_3 \mid x_1, x_2, y_1, \ldots, y_n, \theta\right)}{P\left(x_3 \mid x_1, x_2, y_1, \ldots, y_n\right)} \\ & + \cdots\end{aligned}$$

After using the chain rule for conditional probability the right hand side simplifies to

$$\sum_{x_1, \ldots, x_n, y_1, \ldots, y_n} \pi(x_1, \ldots, x_n, y_1, \ldots, y_n \mid \theta) \ln \frac{\pi\left(x_1, \ldots, x_n \mid y_1, \ldots, y_n, \theta\right)}{P\left(x_1, \ldots, x_n \mid y_1, \ldots, y_n\right)}.$$

Consequently, an alternative way for defining our query set is to define it as a single pair $\{(\mathbf{X}, \mathbf{E})\}$, where $\mathbf{X}$ is the vector of n game's result, and $\mathbf{E}$ is the vector of n dice outcome.

Using the new definition for $\mathcal{X}$, by Equation 14 we have:

$$\sum_{\mathbf{x}, \mathbf{e}} \pi(\mathbf{x}, \mathbf{e} \mid \theta) \ln \frac{\pi\left(\mathbf{x} \mid \mathbf{e}, \theta\right)}{C_{\mathcal{X}}\left(\mathbf{x} \mid \mathbf{e}\right)} = \lambda$$

derive a $\theta = (\alpha_i, \beta_i)$ closed form for central dist...

$$\frac{\pi(\mathbf{x} \mid \mathbf{e}, \theta)}{C_{\mathcal{X}}(\mathbf{x} \mid \mathbf{e})} = \frac{\pi(\theta \mid \mathbf{x}, \mathbf{e})}{C_{\mathcal{X}}(\theta \mid \mathbf{e})}$$

sdfd

$$\sum_{\mathbf{x},\mathbf{e}} \pi(\mathbf{x}, \mathbf{e} \mid \alpha, \beta) \ln \frac{c_{\mathcal{X}}(\beta \mid \mathbf{x}, \mathbf{e})}{c_{\mathcal{X}}(\beta \mid \alpha)}$$

Assume that we decide to simplify our learning task. We choose to consider only the least significant digit of the dice outcome and ignore the other digit. For example, if the dice outcome is 10, we treat it as 0. This approach decreases the number of possible outcomes of the dice, making the learning process simpler. However, it also reduces our ability to predict the game's result accurately.

In order to find a central distribution for this version of our game, we should slightly modify the definition of the query set. While each X_i remains unchanged, we need to represent each E_i as a vector

$$(X_1, X_2, \ldots, X_{i-1}, Y'_1, Y'_2, \ldots, Y'_n),$$

where each Y'_i represents the least significant digit of the dice outcome. We denote this new query set as $\mathcal{X}'$ and the previous query set as $\mathcal{X}$.

Following similar steps as we did for $\mathcal{X}$, we can show that $\mathcal{X}'$ also has a more compact counterpart, which includes a single pair $(\mathbf{X}, \mathbf{E}')$, where $\mathbf{X}$ is the vector $(X_1, X_2, \ldots, X_n)$, and $\mathbf{E}'$ is the vector $(Y'_1, Y'_2, \ldots, Y'_n)$.

explain how the central distribution for this version is different...

When we have a lot of learning data, there is no point in using this simpler version of our learning task. However, when the available data is not enough for the more complex version, the simpler version can still produce relatively meaningful results, while the data is too small for the more complex version.

Is it possible to have a probabilistic inference framework that is able to automatically choose between these two versions of the problem based on the quality and amount of available data? It seems that the central distributing has this ability. For that, we need to use the central distribution for the query set $\mathcal{X} \cup \mathcal{X}'$. This seems reasonable, since when we want our probabilistic inference framework to choose between two versions of our problem, our query set should include random variables of both versions.

Theorem 6. *Let $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$ be a generic parametric model, and $\mathcal{X} = \{X_1, X_2, X_3, \ldots\}$ be a set of random variables in $\mathcal{M}_\Theta$. Let Ω_i, for every i, be a function of the parameter Θ that fully parametrizes X_i, so that for $\omega_i = \Omega_i(\theta)$ we have:*

$$\pi(x_i \mid \theta) = \pi\left(x_i \mid \omega_i\right) \cdot \tag{15}$$

The prior distribution corresponding to the central distribution for $\mathcal{X}$ in $\mathcal{M}_\Theta$ can be characterized by the following fixed point equation:

$$c_\mathcal{X}(\theta) \propto \left(\prod_i c_\mathcal{X}\theta \mid \omega_i \rho(\beta_i \mid \alpha_i)\right)^{\frac{1}{|\mathcal{X}|}}, \tag{16}$$

where

$$\rho(\beta_i \mid \alpha_i) \propto \exp\left\{\sum_{x_i} \pi(x_i \mid \omega_i) \ln c_\mathcal{X}\omega_i \mid x_i\right\}\cdot$$

Proof. First, we show that the set of distributions of each X_i is a convex set. Let $Q^*(x_i) = \alpha Q_1(x_i) + (1-\alpha)Q_2(x_i)$, for some $Q_1, Q_2 \in \mathcal{P}$, $0 \le \alpha \le 1$ and arbitrary i, x_i.

$\mathcal{M}_\Theta$ is a parametric model. Therefore,

$$\begin{aligned} Q^*(x_i) &= \alpha \int \pi(x_i \mid \theta) q_1(\theta) d\theta + (1-\alpha) \int \pi(x_i \mid \theta) q_2(\theta) d\theta \\ &= \int \pi(x_i \mid \theta)(\alpha q_1(\theta) + (1-\alpha) q_2(\theta)) d\theta\cdot \end{aligned}$$

$\alpha q_1(\theta) + (1-\alpha)q_2(\theta)$ is a valid prior distribution and $\mathcal{M}_\Theta$ is a generic parametric model. That implies $Q^* \in \mathcal{P}$.

Thus, by Theorem 5 the central distribution of $\mathcal{X}$ satisfies

$$\sum_i \sum_{x_i} \pi(x_i \mid \theta) \ln \frac{\pi(x_i \mid \theta)}{C_\mathcal{X}(x_i)} = \lambda\cdot \tag{17}$$

Moreover, from Equation 18 and Bayes' theorem it follows

$$\begin{aligned} \sum_i \sum_{x_i} \pi(x_i \mid \theta) \ln \frac{\pi(x_i \mid \theta)}{C_\mathcal{X}(x_i)} &= \sum_i \sum_{x_i} \pi(x_i \mid \omega_i) \ln \frac{\pi(x_i \mid \omega_i)}{C_\mathcal{X}(x_i)} \\ &= \sum_i \sum_{x_i} \pi(x_i \mid \omega_i) \ln \frac{c_\mathcal{X}\omega_i \mid x_i}{c_\mathcal{X}(\omega_i)} \\ &= \sum_i \ln \frac{\rho(\beta_i \mid \alpha_i)}{c_\mathcal{X}(\omega_i)}\cdot \end{aligned}$$

Since each Ω_i is a function of Θ, for $\Omega_i(\theta) = \omega_i$ we have $c_\mathcal{X}\theta, \omega_i = c_\mathcal{X}(\theta)$. Therefore,

$$c_\mathcal{X}(\omega_i) = \frac{c_\mathcal{X}(\theta)}{c_\mathcal{X}\theta \mid \omega_i}\cdot$$

Now we rewrite Equation 20:

$$|\mathcal{X}| \ln c_{\mathcal{X}}(\theta) = \sum_i \ln\left(c_{\mathcal{X}}\theta \mid \omega_i \rho(\beta_i \mid \alpha_i)\right) - \lambda$$

$$c_{\mathcal{X}}(\theta) = \left(e^{-\lambda} \prod_i c_{\mathcal{X}}\theta \mid \omega_i \rho(\beta_i \mid \alpha_i)\right)^{\frac{1}{|\mathcal{X}|}} \qquad \square$$

Theorem 7. *Let $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$ be a generic parametric model, and $\mathcal{X} = \{(X_1, E_1), (X_2, E_2), \ldots\}$ be a set of random variables in $\mathcal{M}_\Theta$. Let Ω_i, for every i, be a function of the parameter Θ that fully parametrizes X_i, so that for $\omega_i = \Omega_i(\theta)$ we have:*

$$\pi(x_i \mid \theta) = \pi\left(x_i \mid \omega_i\right) \cdot \tag{18}$$

The prior distribution corresponding to the central distribution for $\mathcal{X}$ in $\mathcal{M}_\Theta$ can be characterized by the following fixed point equation:

$$c_{\mathcal{X}}(\beta \mid \alpha) \propto \left(\prod_i \frac{c_{\mathcal{X}}(\alpha, \beta \mid \alpha_i, \beta_i)\rho(\beta_i \mid \alpha_i)}{c_{\mathcal{X}}(\alpha \mid \alpha_i)}\right)^{\frac{1}{|\mathcal{X}|}}, \tag{19}$$

where

$$\rho(\beta_i \mid \alpha_i) \propto \exp\left\{\sum_{x_i, e_i} \pi(x_i, e_i \mid \alpha_i, \beta_i) \ln c_{\mathcal{X}}(\beta_i \mid x_i, e_i)\right\}\cdot$$

Proof. First, we show that the set of distributions of each X_i is a convex set. Let $Q^*(x_i \mid e_i) = \alpha Q_1(x_i \mid e_i) + (1-\alpha)Q_2(x_i \mid e_i)$, for some $Q_1, Q_2 \in \mathcal{P}$, $0 \leq \alpha \leq 1$ and arbitrary i, x_i, e_i.

$\mathcal{M}_\Theta$ is a parametric model. Therefore,

$$\begin{aligned} Q^*(x_i \mid e_i) &= \alpha \int \pi(x_i \mid e_i, \beta) q_1(\beta \mid e_i) d\beta + (1-\alpha) \int \pi(x_i \mid \theta) q_2(\theta) d\theta \\ &= \int \pi(x_i \mid \theta)(\alpha q_1(\theta) + (1-\alpha) q_2(\theta)) d\theta \cdot \end{aligned}$$

$\alpha q_1(\theta) + (1-\alpha) q_2(\theta)$ is a valid prior distribution and $\mathcal{M}_\Theta$ is a generic parametric model. That implies $Q^* \in \mathcal{P}$.

Thus, by Theorem 5 the central distribution of $\mathcal{X}$ satisfies

$$\sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{\pi\left(x_i \mid e_i, \theta\right)}{C_{\mathcal{X}}\left(x_i \mid e_i\right)} = \lambda \cdot \tag{20}$$

Moreover, from Equation 18 and Bayes' theorem it follows

$$\begin{aligned}
\lambda &= \sum_i \sum_{x_i,e_i} \pi(x_i, e_i \mid \theta) \ln \frac{\pi\,(x_i \mid e_i, \theta)}{C_{\mathcal{X}}\,(x_i \mid e_i)} \\
&= \sum_i \sum_{x_i,e_i} \pi(x_i, e_i \mid \alpha_i, \beta_i) \ln \frac{\pi(x_i \mid e_i, \beta_i)}{C_{\mathcal{X}}(x_i \mid e_i)} \\
&= \sum_i \sum_{x_i,e_i} \pi(x_i, e_i \mid \alpha_i, \beta_i) \ln \frac{c_{\mathcal{X}}(\beta_i \mid x_i, e_i)}{c_{\mathcal{X}}(\beta_i \mid e_i)} \\
&= \sum_i \ln \frac{\rho(\beta_i \mid \alpha_i)}{c_{\mathcal{X}}(\beta_i \mid \alpha_i)}\cdot
\end{aligned}$$

Since each Ω_i is a function of Θ, for $\Omega_i(\theta) = \omega_i$ we have $c_{\mathcal{X}}\theta, \alpha_i, \beta_i = c_{\mathcal{X}}(\theta)$ $c_{\mathcal{X}}\alpha, \alpha_i = c_{\mathcal{X}}\alpha$. Therefore,

$$c_{\mathcal{X}}(\beta_i \mid \alpha_i) = \frac{c_{\mathcal{X}}(\alpha \mid \alpha_i)}{c_{\mathcal{X}}(\alpha, \beta \mid \alpha_i, \beta_i)} c_{\mathcal{X}}(\beta \mid \alpha)\cdot$$

Now we rewrite Equation 20:

$$|\mathcal{X}| \ln c_{\mathcal{X}}(\beta \mid \alpha) = -\lambda + \sum_i \ln \frac{c_{\mathcal{X}}(\alpha, \beta \mid \alpha_i, \beta_i)\rho(\beta_i \mid \alpha_i)}{c_{\mathcal{X}}(\alpha \mid \alpha_i)}$$

Then:

$$\begin{aligned}
c_{\mathcal{X}}(\beta \mid \alpha) &= \left(e^{-\lambda} \prod_i \frac{c_{\mathcal{X}}(\alpha, \beta \mid \alpha_i, \beta_i)\rho(\beta_i \mid \alpha_i)}{c_{\mathcal{X}}(\alpha \mid \alpha_i)} \right)^{\frac{1}{|\mathcal{X}|}} \\
&= \left(e^{-\lambda} \prod_i \frac{c_{\mathcal{X}}(\beta \mid \alpha)\rho(\beta_i \mid \alpha_i)}{c_{\mathcal{X}}(\beta_i \mid \alpha_i)} \right)^{\frac{1}{|\mathcal{X}|}} \\
&= c_{\mathcal{X}}(\beta \mid \alpha) \left(\prod_i \frac{\rho(\beta_i \mid \alpha_i)}{c_{\mathcal{X}}(\beta_i \mid \alpha_i)} \right)^{\frac{1}{|\mathcal{X}|}} \\
&= c_{\mathcal{X}}(\beta \mid \alpha) \left(e^{-\lambda} \prod_i \frac{\rho(\alpha_i, \beta_i) c_{\mathcal{X}}(\alpha_i)}{c_{\mathcal{X}}(\alpha_i, \beta_i)\rho(\alpha_i)} \right)^{\frac{1}{|\mathcal{X}|}} \qquad \square
\end{aligned}$$

References

[1] José M. Bernardo. Reference analysis. In D.K. Dey and C.R. Rao, editors, *Bayesian Thinking*, volume 25 of *Handbook of Statistics*, pages 17–90. Elsevier, 2005.

[2] Ding-Zhu Du. *Minimax and Its Applications*, pages 339–367. Springer US, Boston, MA, 1995.